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COMPUTER SCIENCE

PROJECT FILE

Employee Management

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SHARJAH INDIAN SCHOOL

(AFFILIATED TO CENTRAL BOARD OF SECONDARY EDUCATION, NEW DELHI)

Certificate

AISSCE Roll No : _____

Certified that the work entered in this journal on the Project

Employee Management is the bona fide work of

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Acknowledgement

I would like to thank Almighty, my parents and my teachers for giving me this opportunity and for giving me great encouragement to do this Project work. I would also like to thank my teammate for supporting and helping me complete this Project successfully.

Name of the Student

Signature

About Python

Python is a widely-used, interpreted, object-oriented, and high-level programming language with dynamic semantics, used for general-purpose programming. It was created by Guido van Rossum, and first released on February 20, 1991.

Python can be easy to pick up whether you're a first time programmer or you're experienced with other languages. Python is developed under an OSI-approved open source license, making it freely usable and distributable, even for commercial use.

About the Project

Employee Management

This project on 'Employee Management' has been developed in Python. This program consists of ten options for Admins and two options for Employees as follows:

Admin:

- To add a new employee
- To modify employee details
- To remove an employee
- To view employee details
- To sort employees by department
- To count the total number of employees working
- To manage bonuses
- To view employee card of an employee
- To change admin login details
- Log out

Employee:

- To view your employee card by entering employee number
- Log out

The program uses SQL Connectivity to enable permanent storage of data.

i. Modules

Modules are files containing Python codes; which makes the program easy to understand, test & maintain

The Modules used in this Project are:

1. mysql.connector – used for creating a connection with MySQL
2. sys - provides functions and variables used to manipulate different parts of the Python runtime environment
3. time - provides various time-related functions
4. tabulate – used to convert to tabular form
5. datetime - supplies classes for manipulating dates and times

ii. Classes

Python is an object oriented programmeng language. OOP is a method of structuring a program by bundling related properties and behaviors into individual **objects**. Everything in Python is an object, with its properties and methods. A Class is like an object constructor, or a "blueprint" for creating objects.

iii. Functions

Functions are subprograms that perform a specific task. They provide a systematic way of problem solving by dividing the given problem into several sub-programs, finding their individual solutions and integrating the solutions of individual problems to solve the original problem.

Functions used in this project are :

1. def admin_access():To access the admin menu
2. def addrecord(): To add records
3. def delrecord(): To delete a record
4. def displayall(): To display details of all employees
5. def displayno(): To display details of an employee by employee no
6. def display(): Menu to select displayall() or displayno()

7. def displaydept(): Displays employees based on selected department
8. def updatename(): To update name of an employee
9. def updatesal(): To update salary of an employee
10. def updatedep(): To update department of an employee
11. def updatestat(): To update payment status of an employee
12. def updatecontact(): To update contact number of an employee
13. def updates(): Menu to choose what to update
14. def count(): To count the total number of employees in the company
15. def bonus(): To give bonuses to employees
16. def logindetails(): To change the login details
17. def employee_card(): Displays card of the employee by entering employee number

iv. Files/SQL Tables

Files enable permanent storage of the records needed in the Project.

SQL Table is used in the Project because (write advantages of Binary File)

Files used in this project are :

1. Empmanagement – Database
2. Employee – Table to store data

User Manual

This user manual will help the user in executing this software.

- Hardware Requirements:
  - X86 64-bit CPU (Intel / AMD Architecture)
  - 4 GB RAM
  - 5 GB free disk space
- Software Requirements
  - Modern Operating System
    - Windows 7 above
    - Mac OS X 10.11 or higher, 64 bit
    - Linux : RHEL 6/7, 64 bit
  - Python 3.8 or above
- As the program is very simple the user just needs fundamental knowledge on computers.
- The installation of the program is very simple. Just insert the CD and execute the .py file using Python.

Source Code

```
import mysql.connector as ms
import sys
import time
from tabulate import tabulate
from datetime import datetime

################ FUNCTION - ADMIN ################


def admin_access():
    print()
    global aaccess
    aaccess = False
    ch="Y"
    while ch in 'Yy':
        usern = input('Enter username  : ')
        pasw = input('Enter password  : ')
        if usern == user and pasw == passwd:
            aaccess = True
        if aaccess:
            print()
            color.write("Access granted : )\n","STRING")
            print("Opening menus....")
            time.sleep(2)
            print("\n")
            break
        else:
            print()
            color.write("Incorrect username or password","COMMENT")
            print()
```

```
            ch=input("Do you wish to try again? (Y/N): ")
            print("\n")


############### FUNCTION - ADD ###############


def addrecord():
    chn='Y'
    while chn in 'Yy':
        a=input("Enter employee number: ")
        try:
            int(a)
            chn='N'
        except:
            color.write("Invalid value.. Please try again","KEYWORD")
            time.sleep(1.5)
            print("\n")
    print()
    b=input("Enter Name of the employee: ")
    print()
    while True:
        c=input('''Departments
|    1. Transport     |
|    2. Office        |
|    3. Cleaning      |
|    4. Maintenance   |
|    5. Security      |
|    6. Sales         |
|    7. Accounts      |
Choose your department: ''')
        if c == '1':
            y = 'Transport'
```

```
            break
        elif c == '2':
            y = 'Office'
            break
        elif c == '3':
            y = 'Cleaning'
            break
        elif c == '4':
            y = 'Maintenance'
            break
        elif c == '5':
            y = 'Security'
            break
        elif c == '6':
            y = 'Sales'
            break
        elif c == '7':
            y = 'Accounts'
            break
        else:
            color.write("Invalid value.. Please try again","KEYWORD")
            time.sleep(1.5)
            print("\n")
    print()
    c=y
    cho='Y'
    while cho in 'Yy':
        d=input("Enter salary: ")
        try:
            int(d)
            cho='N'
        except:
            color.write("Invalid value.. Please try again","KEYWORD")
```

```
            time.sleep(1.5)
            print("\n")

    print()
    while True:
        e=input("Enter Payment status (P->Paid N->Not paid): ")
        if e.upper() in 'P':
            e='Paid'
            break
        elif e.upper() in 'N':
            e='Not Paid'
            break
        else:
            e='Not available'
            color.write("Invalid value.. Please try again","KEYWORD")
            time.sleep(1.5)
            print("\n")
    print()
    f="None"
    g=input("Enter contact no: ")
    try:
        mc.execute("insert into employee values
('{}','{}','{}',{},'{}','{}','{}')".format(a,b,c,d,e,f,g))
        mydb.commit()
        print()
        color.write("Record Saved","STRING")
        time.sleep(1.5)
    except:
        print()
        print("Record not Saved")
        mydb.rollback()
    print("\n")
```

```
############### FUNCTION - DELETE ###############

def delrecord():
    try:
        print()
        no=int(input("Enter employee no to be removed: "))
        mc.execute("delete from employee where empid={}".format(no))
        c=mc.rowcount
        if c>0:
            co='Y'
            while co in 'Yy':
                ans=input("Are you sure you want to remove the employee?(Y/N):
")
                if ans=='y' or ans=='Y':
                    co='N'
                    mydb.commit()
                    print()
                    color.write("The employee has been removed
successfully\n","STRING")
                    print()
                    time.sleep(1)
                elif ans=='n' or ans=='N':
                    co='N'
                    mydb.rollback()
                    color.write("Exiting back to menu...\n","BUILTIN")
                    print()
                    time.sleep(1)
                else:
                    color.write("Invalid choice... Please try
again\n","COMMENT")
                    print()
                    time.sleep(1)
        else:
```

```
            print()
            color.write("No records found with matching employee
number","KEYWORD")
            time.sleep(1)
            print("\n")
    except:
        print()
        print('An unexpected error occured :(')
        print("Record not Saved")
        mydb.rollback()
        print("\n")


############### FUNCTION - DISPLAY ###############


def displayall():
    try:
        mc.execute("select * from employee")
        print("\n")

color.write(tabulate(mc,headers=['EmpNo','EmpName','Department','Salary','Stat
us','Bonus','Contact no'],tablefmt='psql'),"SYNC")
        print("\n")
        if mc.rowcount==-1:
            color.write("No records found\n","KEYWORD")
            print("\n")
    except:
        print("Something went wrong")

def displayno():
    try:
        x=int(input("Enter the employee number: "))
        mc.execute("select * from employee where empid={}".format(x))
```

```
        print("\n")

color.write(tabulate(mc,headers=['EmpNo','EmpName','Department','Salary','Stat
us','Bonus','Contact no'],tablefmt='psql'),"SYNC")
        print("\n")
        if mc.rowcount==-1:
            color.write("No records found with matching emploee
number\n","KEYWORD")
            print("\n")
    except:
        print("Something went wrong")
        print("\n")

def display():
    while True:
        print("\n|   1. Display details of all employees          |")
        print("|   2. Display details of employee with emp no   |")
        print("|   0. Use any other number to exit              |")
        ch=int(input("\nEnter a choice: "))
        if ch==1:
            displayall()
        elif ch==2:
            displayno()
        else:
            color.write("Exiting....\n","BUILTIN")
            time.sleep(2)
            print("-"*48)
            print("\n")
            break

def displaydept():
    try:
        print("Departments:")
        print("      |    1. Transport     | ")
```

```
        print("      |    2. Office        | ")
        print("      |    3. Cleaning      | ")
        print("      |    4. Maintenance   | ")
        print("      |    5. Security      | ")
        print("      |    6. Sales         | ")
        print("      |    7. Accounts      | ")
        x=input("Enter a choice: ")
        if x=='1':
            y="Transport"
        elif x=='2':
            y="Office"
        elif x=='3':
            y="Cleaning"
        elif x=='4':
            y="Maintenance"
        elif x=='5':
            y="Security"
        elif x=='6':
            y="Sales"
        elif x=='7':
            y="Accounts"
        else:
            color.write("Invalid choice","COMMENT")
            time.sleep(1.5)
            print("\n")
        mc.execute("select * from employee where department='{}'".format(y))
        print("\n")

color.write(tabulate(mc,headers=['EmpID','EmpName','Department','Salary','Stat
us','Bonus','Contact no'],tablefmt='psql'),"SYNC")
        print("\n")
        if mc.rowcount==-1:
            color.write("No records found\n","KEYWORD")
            print("\n")
```

```
    except:
        print("Please try again")
        print("\n")


############### FUNCTION - UPDATE ###############


def updatename():
    print()
    no=int(input("Enter employee no: "))
    mc.execute("select * from employee where empid={}".format(no))
    d=mc.fetchall()
    if mc.rowcount<=0:
        color.write("No records found with matching employee
number\n","COMMENT")
        print()
        time.sleep(1.2)
    else:
        for i in d:
            print("Current employee: ",i[1])
        co='Y'
        while co in 'Yy':
            ans=input("Are you sure you want to update?(Y/N): ")
            if ans=='y' or ans=='Y':
                co='N'
                x=input("Enter the new employee: ")
                mc.execute("update employee set empname='{}' where
empid={}".format(x,no))
                mydb.commit()
                color.write("Record updated successfully\n","STRING")
                print()
```

```
                time.sleep(1)
            elif ans=='n' or ans=='N':
                co='N'
                color.write("Exiting back to menu...\n","BUILTIN")
                print()
                time.sleep(1)
            else:
                color.write("Invalid choice... Please try again\n","COMMENT")
                print()
                time.sleep(1)

def updatesal():
    print()
    no=int(input("Enter employee no: "))
    mc.execute("select * from employee where empid={}".format(no))
    d=mc.fetchall()
    if mc.rowcount<=0:
        color.write("No records found with matching employee
number\n","COMMENT")
        print()
        time.sleep(1)
    else:
        for i in d:
            print("Employee name: ",i[1])
            print("Current salary: ",i[3])
        co='Y'
        while co in 'Yy':
            ans=input("Are you sure you want to update?(Y/N): ")
            if ans=='y' or ans=='Y':
                co='N'
                sal=int(input("Enter new salary: "))
                mc.execute("update employee set salary={} where
empid={}".format(sal,no))
                mydb.commit()
```

```
                color.write("Record updated successfully\n","STRING")
                print()
                time.sleep(1)
            elif ans=='n' or ans=='N':
                co='N'
                color.write("Exiting back to menu...\n","BUILTIN")
                print()
                time.sleep(1)
            else:
                color.write("Invalid choice... Please try again\n","COMMENT")
                print()
                time.sleep(1)

def updatedep():
    print()
    no=int(input("Enter employee no: "))
    mc.execute("select * from employee where empid={}".format(no))
    d=mc.fetchall()
    if mc.rowcount<=0:
        color.write("No records found with matching employee
number\n","COMMENT")
        print()
        time.sleep(1)
    else:
        for i in d:
            print("Employee name: ",i[1])
            print("Current department: ",i[2])
        co='Y'
        while co in 'Yy':
            ans=input("Are you sure you want to update?(Y/N): ")
            if ans=='y' or ans=='Y':
                co='N'
                cho='Y'
```

```
                while cho in 'Yy':
                    c=input('''\nDepartments
|    1. Transport     |
|    2. Office        |
|    3. Cleaning      |
|    4. Maintenance   |
|    5. Security      |
|    6. Sales         |
|    7. Accounts      |\n
Choose your department: ''')
                    if c == '1':
                        y = 'Transport'
                        cho='N'
                    elif c == '2':
                        y = 'Office'
                        cho='N'
                    elif c == '3':
                        y = 'Cleaning'
                        cho='N'
                    elif c == '4':
                        y = 'Maintenance'
                        cho='N'
                    elif c == '5':
                        y = 'Security'
                        cho='N'
                    elif c == '6':
                        y = 'Sales'
                        cho='N'
                    elif c == '7':
                        y = 'Accounts'
                        cho='N'
                    else:
                        color.write("Invalid choice... Please try
```

```
again\n","COMMENT")
                        time.sleep(1.5)
                        print("\n")

                mc.execute("update employee set department='{}' where
empid={}".format(y,no))
                mydb.commit()
                color.write("Record updated successfully\n","STRING")
                print()
                time.sleep(1)
            elif ans=='n' or ans=='N':
                co='N'
                color.write("Exiting back to menu...\n","BUILTIN")
                print()
                time.sleep(1)
            else:
                color.write("Invalid choice... Please try again\n","COMMENT")
                print()
                time.sleep(1)

def updatestat():
    print()
    no=int(input("Enter employee no: "))
    mc.execute("select * from employee where empid={}".format(no))
    d=mc.fetchall()
    if mc.rowcount<=0:
        color.write("No records found with matching employee
number\n","COMMENT")
        print()
        time.sleep(1)
    else:
        for i in d:
            print("Employee name: ",i[1])
            print("Current status: ",i[4])
```

```
        co='Y'
        while co in 'Yy':
            ans=input("Are you sure you want to update?(Y/N): ")
            if ans=='y' or ans=='Y':
                co='N'
                cc='Y'
                while cc in 'Yy':
                    stat=input("Enter new payment status (P->Paid / N->Not
paid) : ")
                    if stat.upper() in 'P':
                        cc='N'
                        stat='Paid'
                    elif stat.upper() in 'N':
                        cc='N'
                        stat='Not Paid'
                    else:
                        print("Invalid choice... Please try again\n")
                        time.sleep(1)
                mc.execute("update employee set status='{}' where
empid={}".format(stat,no))
                mydb.commit()
                color.write("Record updated successfully\n","STRING")
                print()
                time.sleep(1)
            elif ans=='n' or ans=='N':
                co='N'
                color.write("Exiting back to menu...\n","BUILTIN")
                print()
                time.sleep(1)
            else:
                color.write("Invalid choice... Please try again\n","COMMENT")
                print()
                time.sleep(1)
```

```
def updatecontact():
    print()
    no=int(input("Enter employee no: "))
    mc.execute("select * from employee where empid={}".format(no))
    d=mc.fetchall()
    if mc.rowcount<=0:
        color.write("No records found with matching employee
number\n","COMMENT")
        print()
        time.sleep(1.2)
    else:
        for i in d:
            print("Employee name: ",i[1])
            print("Current contact number: ",i[6])
        co='Y'
        while co in 'Yy':
            ans=input("Are you sure you want to update?(Y/N): ")
            if ans=='y' or ans=='Y':
                co='N'
                cont=input("Enter the new contact number: ")
                mc.execute("update employee set contact='{}' where
empid={}".format(cont,no))
                mydb.commit()
                color.write("Record updated successfully\n","STRING")
                print()
                time.sleep(1)
            elif ans=='n' or ans=='N':
                co='N'
                color.write("Exiting back to menu...\n","BUILTIN")
                print()
                time.sleep(1)
            else:
                color.write("Invalid choice... Please try again\n","COMMENT")
                print()
```

```
                time.sleep(1)

def updates():
    c='Y'
    while c in 'Yy':
        passw=input("Enter security password to continue: ")
        if passw=="1234":
            color.write("\nPassword correct.... opening menus\n","STRING")
            print()
            time.sleep(1.5)
            while True:
                update_op = input('''
                |     1. Update name           |
                |     2. Update salary         |
                |     3. Update department     |
                |     4. Update status         |
                |     5. Update contact no     |
                |     0. Exit                  |
                Enter a choice : ''')
                if update_op == '1':
                    updatename()
                elif update_op == '2':
                    updatesal()
                elif update_op == '3':
                    updatedep()
                elif update_op == '4':
                    updatestat()
                elif update_op == '5':
                    updatecontact()
                else:
                    color.write("Exiting...\n","BUILTIN")
                    time.sleep(2)
                    print("-"*48)
```

```
                    print("\n")
                    break
            break
        else:
            print()
            color.write("Invalid password\n","COMMENT")
            print()
            c=input("Do you want to try again? (Y/N): ")
            if c=='N':
                color.write("Exiting...\n","BUILTIN")
                time.sleep(2)
                print("\n")


############### FUNCTION - COUNT ###############


def count():
    print()
    mc.execute("select * from employee")
    data=mc.fetchall()
    count=0
    if mc.rowcount<=0:
    color.write("No employees found\n","KEYWORD")
        print()
        time.sleep(1.2)
    else:
        for i in data:
            count=count+1
        color.write("Total number of employees in the company: ","SYNC")
        color.write(str(count),"SYNC")
        print("\n")
        time.sleep(1)
```

```
############### FUNCTION - BONUS ###############


def bonus():
    c='Y'
    while c in 'Yy':
        print('''\n|   1. Set employee of the month       |
|   2. Current employee of the month   |
|   3. Remove employee of the month    |
|   4. Salary bonus                    |
|   0. Exit                            |\n''')
        ch=int(input("Enter a choice: "))
        if ch==1:
            print()
            no=int(input("Enter employee no: "))
            mc.execute("select * from employee where empid={}".format(no))
            d=mc.fetchall()
            if mc.rowcount<=0:
                color.write("No records found with matching employee
number\n","COMMENT")
                print()
                time.sleep(1)
            else:
                for i in d:
                    print("Employee name: ",i[1])
                x='Employee of the month'
                y='None'
                mc.execute("update employee set bonus='{}'".format(y))
                mc.execute("update employee set bonus='{}' where
empid={}".format(x,no))
                color.write("Successfully updated bonus\n","STRING")
                print()
```

```
                time.sleep(1)
                mydb.commit()

        elif ch==2:
            print()
            x='Employee of the month'
            mc.execute("select * from employee where bonus='{}'".format(x))
            data=mc.fetchall()
            if mc.rowcount<=0:
                color.write("No employee of the month found\n","KEYWORD")
                print()
                time.sleep(1)
            else:
                for i in data:
                    print("Current employee of the month: ",i[1])
                    print("\n")
                    time.sleep(1)
        elif ch==3:
            no=input("Enter employee no: ")
            mc.execute("select * from employee where empid={}".format(no))
            d=mc.fetchall()
            found=True
            if mc.rowcount<=0:
                color.write("No records found with matching employee
number\n","COMMENT")
                time.sleep(1)
            else:
                for i in d:
                    if i[5]=="Employee of the month":
                        print("Employee name: ",i[1])
                        found=True
                    else:
                        print("Employee name: ",i[1])
```

```
                        print(i[1],"is not the current Employee of the month")
                        time.sleep(1)
                        print("\n")
                        found=False
                if found==True:
                    y='None'
                    mc.execute("update employee set bonus='{}' where
empid={}".format(y,no))
                    color.write("Successfully removed employee of the
month\n","STRING")
                    print()
                    time.sleep(1)
                    mydb.commit()

        elif ch==4:
            print()
            no=int(input("Enter employee no to give salary bonus: "))
            mc.execute("select * from employee where empid={}".format(no))
            d=mc.fetchall()
            if mc.rowcount<=0:
                print()
                color.write("No records found with matching employee
number\n","COMMENT")
                print()
                time.sleep(1)
            else:
                for i in d:
                    print("Employee name: ",i[1])
                    print("Current salary: ",i[3])
                co='Y'
                while co in 'Yy':
                    co='N'
                    s=float(input("Enter new percentage to increase: "))
                    sal=s/100
```

```
                    mc.execute("update employee set salary=salary+salary*{}
where empid={}".format(sal,no))
                    mydb.commit()
                    color.write("Salary bonus added successfully\n","STRING")
                    print()
                    time.sleep(1)

        else:
            color.write("Exiting...\n","BUILTIN")
            time.sleep(2)
            print("-"*48)
            print("\n")
            break

############### FUNCTION - LOGIN ###############


def logindetails():
    global user
    global passwd
    c='Y'
    while c in 'Yy':
        x=input("Enter current username: ")
        y=input("Enter current password: ")
        if x==user and y==passwd:
            c='N'
            print()
            color.write("Access granted\n","STRING")
            print()
            u=input("Enter new username: ")
            p=input("Enter new password: ")
            user=u
            passwd=p
```

```
            color.write("Successfully changed\n","STRING")
            print()
        else:
            color.write("\nIncorrect username or password\n","COMMENT")
            print()
            c=input("Do you wish to try again? (Y/N): ")
            if c=='N':
                color.write("Exiting...\n","BUILTIN")
                time.sleep(2)
                print("\n")


############### FUNCTION - CARD ###############


def employee_card():
    print()
    id=input('Enter employee no : ')
    print("\n")
    mc.execute("select * from employee where empid='{}'".format(id))
    data=mc.fetchall()
    now=datetime.now()
    month=now.strftime('%B')
    year=now.strftime('%Y')
    if mc.rowcount<=0:
        color.write("-"*54,"COMMENT")
        print()
        color.write("!!! No records found with matching employee number
!!!\n","COMMENT")
        color.write("-"*54,"COMMENT")
        print("\n")
        time.sleep(1)
    else:
```

```
        print('-'*70)
        print('RECRUITCARE EMPLOYMENT AGNECY'.center(70))
        print('Shrajah, UAE'.center(70))
        print('email: contact@Recruitcareagency.ae   phone: 06 598
1203'.center(70))
        print()
        print('\t\t    EMPLOYEE CARD -',month,year)
        print()
        print('ID:',data[0][0])
        print()
        print('NAME:',data[0][1])
        print()
        print('DEPARTMENT:',data[0][2])
        print()
        print('SALARY:',data[0][3])
        print()
        print('SALARY STATUS:',data[0][4])
        print()
        print('CONTACT NO:',data[0][6])
        print()
        print('Generated on:',datetime.now())
        print()
        if data[0][5]=="Employee of the month":
            print("!!",data[0][1],"is the current Employee of the month !!")
            print()
        print('THANK YOU'.center(70))
        print('-'*70)
        print("\n")
        color.write("Please press enter to continue\n","KEYWORD")
        input()
        print()

    continu=input('''
```

```
1. Use another employee id
0. Log out / return to menu
Enter a choice : ''')
    if continu not in '1':
        color.write("Returning....\n","BUILTIN")
        print("\n")
        time.sleep(2)
    else:
        employee_card()


############### MAIN PROGRAM ###############


mydb=ms.connect(host="localhost",user="root",passwd='1234',database='EmpManage
ment')

if not mydb.is_connected():
    print("Connection failed")

mc=mydb.cursor()

try:
    color=sys.stdout.shell
except AttributeError:
    raise RuntimeError("Use Python IDLE to continue")

user="admin"
passwd="admin"

cho='Y'
while cho in 'Yy':
    print(' _____________________________________________________ '.center(90))
```

```
    print('|                                                    |'.center(90))
    print('|    WELCOME TO RECRUITCARE EMPLOYEMENT AGENCY    |'.center(90))
    print('|____________________________________________________|'.center(90))
    print("\n")
    while True:
        print("|========LOGIN========|")
        print('''|                     |
|     1. Admin        |
|     2. Employee     |
|_____________________|

3. Exit program''')
        print("")
        ch = input('Enter your choice     : ')
        if ch == '1':
            admin_access()
            if aaccess == True:
                i='Y'
                while i in 'Yy':
                    op = input(''' ____________________________________
|                                    |
|     1. Add employee                |
|     2. Update employee details     |
|     3. Remove employee             |
|     4. View employee details       |
|     5. Sort employee by department |
|     6. Count the no of employees   |
|     7. Manage Bonuses              |
|     8. View employee card          |
|     9. Change login details        |
|     0. Log out                     |
|____________________________________|
```

```
Enter a choice : ''')
                    if op == '1':
                        addrecord()
                    elif op == '2':
                        updates()
                    elif op == '3':
                        delrecord()
                    elif op == '4':
                        display()
                    elif op == '5':
                        displaydept()
                    elif op == '6':
                        count()
                    elif op == '7':
                        bonus()
                    elif op == '8':
                        employee_card()
                    elif op == '9':
                        logindetails()
                    else:
                        color.write("Logging out....\n","BUILTIN")
                        time.sleep(2)
                        print("\n")
                        break
                    cn='Y'
                    while cn in 'Yy':
                        continu=input('Do you wish to continue as admin? (Y/N)
: ')
                        if continu in 'Yy':
                            cn='N'
                            print("\n")
                        elif continu in 'Nn':
                            cn='N'
```

```
                    i='N'
                    color.write("Logging out....\n","BUILTIN")
                    time.sleep(2)
                    print("\n")
                    break
                elif continu not in 'Yy' and continu not in 'Nn':
                    color.write("Invalid choice\n","COMMENT")
                    color.write("Please try again...\n","KEYWORD")
                    print()
                    time.sleep(1)

        elif aaccess == False:
            color.write("Exiting....\n","BUILTIN")
            time.sleep(2)
            print("\n")

    elif ch == '2':
        employee_card()

    elif ch == '3':
        color.write("Exiting....\n","BUILTIN")
        time.sleep(2)
        print("-"*48)
        print("\n")
        cho='N'
        break

    else:
        color.write("Invalid choice.... Please try again\n","COMMENT")
        print()
        time.sleep(1.
```

OUTPUT SCREEN

```
 _____________________________________________________
|                                                     |
|    WELCOME TO RECRUITCARE EMPLOYEMENT AGENCY        |
|_____________________________________________________|


|========LOGIN========|
|                     |
|     1. Admin        |
|     2. Employee     |
|_____________________|

3. Exit program

Enter your choice     : 1

Enter username   : admin
Enter password   : admin

Access granted : )
Opening menus....


 _____________________________________
|                                     |
|     1. Add employee                 |
|     2. Update employee details      |
|     3. Remove employee              |
|     4. View employee details        |
|     5. Sort employee by department  |
|     6. Count the no of employees    |
|     7. Manage Bonuses               |
|     8. View employee card           |
|     9. Change login details         |
|     0. Log out                      |
|_____________________________________|

Enter a choice : 1
Enter employee number: 3020

Enter Name of the employee: Mike
```

```
Departments
|     1. Transport      |
|     2. Office         |
|     3. Cleaning       |
|     4. Maintenance    |
|     5. Security       |
|     6. Sales          |
|     7. Accounts       |
Choose your department: 4

Enter salary: 15000

Enter Payment status (P->Paid N->Not paid): P

Enter contact no: 050-385-6912

Record Saved

Do you wish to continue as admin? (Y/N) : y


 _____________________________________________
|                                             |
|      1. Add employee                        |
|      2. Update employee details             |
|      3. Remove employee                     |
|      4. View employee details               |
|      5. Sort employee by department         |
|      6. Count the no of employees           |
|      7. Manage Bonuses                      |
|      8. View employee card                  |
|      9. Change login details                |
|      0. Log out                             |
|_____________________________________________|

Enter a choice : 4

|    1. Display details of all employees          |
|    2. Display details of employee with emp no   |
|    0. Use any other number to exit              |

Enter a choice: 1
```

```
+---------+-------------+--------------+----------+----------+-----------------------+--------------+
|   EmpNo | EmpName     | Department   |   Salary | Status   | Bonus                 | Contact no   |
|---------+-------------+--------------+----------+----------+-----------------------+--------------|
|    1010 | John Parker | Sales        |    20000 | Paid     | None                  | 050-539-2245 |
|    3326 | Chris       | Office       |    18000 | Paid     | None                  | 050-426-3544 |
|    1026 | Harish      | Cleaning     |    12000 | Not Paid | None                  | 050-134-4252 |
|    3820 | Ritesh      | Accounts     |    20000 | Paid     | None                  | 050-454-0012 |
|    2302 | Harold      | Transport    |    10000 | Paid     | None                  | 050-973-5466 |
|    3370 | Sajit       | Maintenance  |    15000 | Not Paid | None                  | 050-385-3895 |
|    4030 | Tom         | Security     |    10000 | Not Paid | Employee of the month | 050-545-0834 |
|    2890 | Gordon      | Office       |    19000 | Paid     | None                  | 050-623-3541 |
|    3020 | Mike        | Maintenance  |    15000 | Paid     | None                  | 050-385-6912 |
+---------+-------------+--------------+----------+----------+-----------------------+--------------+


|   1. Display details of all employees          |
|   2. Display details of employee with emp no   |
|   0. Use any other number to exit              |

Enter a choice: 2
Enter the employee number: 3020


+---------+-----------+--------------+----------+----------+---------+--------------+
|   EmpNo | EmpName   | Department   |   Salary | Status   | Bonus   | Contact no   |
|---------+-----------+--------------+----------+----------+---------+--------------|
|    3020 | Mike      | Maintenance  |    15000 | Paid     | None    | 050-385-6912 |
+---------+-----------+--------------+----------+----------+---------+--------------+


|   1. Display details of all employees          |
|   2. Display details of employee with emp no   |
|   0. Use any other number to exit              |

Enter a choice: 0
Exiting....
------------------------------------------------


Do you wish to continue as admin? (Y/N) : y


 _____________________________________________
|                                             |
|     1. Add employee                         |
|     2. Update employee details              |
|     3. Remove employee                      |
|     4. View employee details                |
|     5. Sort employee by department          |
|     6. Count the no of employees            |
|     7. Manage Bonuses                       |
|     8. View employee card                   |
|     9. Change login details                 |
|     0. Log out                              |
|_____________________________________________|
```

```
Enter a choice : 2
Enter security password to continue: 123

Invalid password

Do you want to try again? (Y/N): y
Enter security password to continue: 1234

Password correct.... opening menus


                |     1. Update name          |
                |     2. Update salary        |
                |     3. Update department    |
                |     4. Update status        |
                |     5. Update contact no    |
                |     0. Exit                 |
                Enter a choice : 1

Enter employee no: 3020
Current employee:  Mike
Are you sure you want to update?(Y/N): y
Enter the new employee: Nick
Record updated successfully


                |     1. Update name          |
                |     2. Update salary        |
                |     3. Update department    |
                |     4. Update status        |
                |     5. Update contact no    |
                |     0. Exit                 |
                Enter a choice : 3

Enter employee no: 3020
Employee name:  Nick
Current department:  Maintenance
Are you sure you want to update?(Y/N): y

Departments
|    1. Transport     |
|    2. Office        |
|    3. Cleaning      |
|    4. Maintenance   |
|    5. Security      |
|    6. Sales         |
|    7. Accounts      |
```

```
Choose your department: 5
Record updated successfully


                |     1. Update name           |
                |     2. Update salary         |
                |     3. Update department     |
                |     4. Update status         |
                |     5. Update contact no     |
                |     0. Exit                  |
                Enter a choice : 4

Enter employee no: 3020
Employee name:  Nick
Current status:  Paid
Are you sure you want to update?(Y/N): y
Enter new payment status (P->Paid / N->Not paid) : n
Record updated successfully


                |     1. Update name           |
                |     2. Update salary         |
                |     3. Update department     |
                |     4. Update status         |
                |     5. Update contact no     |
                |     0. Exit                  |
                Enter a choice : 0
Exiting...
------------------------------------------------


Do you wish to continue as admin? (Y/N) : y


 _____________________________________________
|                                             |
|     1. Add employee                         |
|     2. Update employee details              |
|     3. Remove employee                      |
|     4. View employee details                |
|     5. Sort employee by department          |
|     6. Count the no of employees            |
|     7. Manage Bonuses                       |
|     8. View employee card                   |
|     9. Change login details                 |
|     0. Log out                              |
|_____________________________________________|

Enter a choice : 4
```

```
|    1. Display details of all employees          |
|    2. Display details of employee with emp no   |
|    0. Use any other number to exit              |

Enter a choice: 1
```

EmpNo	EmpName	Department	Salary	Status	Bonus	Contact no
1010	John Parker	Sales	20000	Paid	None	050-539-2245
3326	Chris	Office	18000	Paid	None	050-426-3544
1026	Harish	Cleaning	12000	Not Paid	None	050-134-4252
3820	Ritesh	Accounts	20000	Paid	None	050-454-0012
2302	Harold	Transport	10000	Paid	None	050-973-5466
3370	Sajit	Maintenance	15000	Not Paid	None	050-385-3895
4030	Tom	Security	10000	Not Paid	Employee of the month	050-545-0834
2890	Gordon	Office	19000	Paid	None	050-623-3541
3020	Nick	Security	15000	Not Paid	None	050-385-6912

```
|    1. Display details of all employees          |
|    2. Display details of employee with emp no   |
|    0. Use any other number to exit              |

Enter a choice: 0
Exiting....
-------------------------------------------------


Do you wish to continue as admin? (Y/N) : y


 _____________________________________________
|                                             |
|     1. Add employee                         |
|     2. Update employee details              |
|     3. Remove employee                      |
|     4. View employee details                |
|     5. Sort employee by department          |
|     6. Count the no of employees            |
|     7. Manage Bonuses                       |
|     8. View employee card                   |
|     9. Change login details                 |
|     0. Log out                              |
|_____________________________________________|

Enter a choice : 3

Enter employee no to be removed: 3020
Are you sure you want to remove the employee?(Y/N): y

The employee has been removed successfully
```

```
Do you wish to continue as admin? (Y/N) : y


 _____________________________________________
|                                             |
|     1. Add employee                         |
|     2. Update employee details              |
|     3. Remove employee                      |
|     4. View employee details                |
|     5. Sort employee by department          |
|     6. Count the no of employees            |
|     7. Manage Bonuses                       |
|     8. View employee card                   |
|     9. Change login details                 |
|     0. Log out                              |
|_____________________________________________|

Enter a choice : 5
Departments:
      |    1. Transport     |
      |    2. Office        |
      |    3. Cleaning      |
      |    4. Maintenance   |
      |    5. Security      |
      |    6. Sales         |
      |    7. Accounts      |
Enter a choice: 2


+---------+-----------+--------------+----------+----------+---------+--------------+
|   EmpID | EmpName   | Department   |   Salary | Status   | Bonus   | Contact no   |
|---------+-----------+--------------+----------+----------+---------+--------------|
|    3326 | Chris     | Office       |    18000 | Paid     | None    | 050-426-3544 |
|    2890 | Gordon    | Office       |    19000 | Paid     | None    | 050-623-3541 |
+---------+-----------+--------------+----------+----------+---------+--------------+

Do you wish to continue as admin? (Y/N) : y


 _____________________________________________
|                                             |
|     1. Add employee                         |
|     2. Update employee details              |
|     3. Remove employee                      |
|     4. View employee details                |
|     5. Sort employee by department          |
|     6. Count the no of employees            |
|     7. Manage Bonuses                       |
|     8. View employee card                   |
|     9. Change login details                 |
|     0. Log out                              |
|_____________________________________________|
```

```
Enter a choice : 6

Total number of employees in the company: 8

Do you wish to continue as admin? (Y/N) : y


 _____________________________________
|                                     |
|     1. Add employee                 |
|     2. Update employee details      |
|     3. Remove employee              |
|     4. View employee details        |
|     5. Sort employee by department  |
|     6. Count the no of employees    |
|     7. Manage Bonuses               |
|     8. View employee card           |
|     9. Change login details         |
|     0. Log out                      |
|_____________________________________|

Enter a choice : 7

|   1. Set employee of the month      |
|   2. Current employee of the month  |
|   3. Remove employee of the month   |
|   4. Salary bonus                   |
|   0. Exit                           |

Enter a choice: 2

Current employee of the month:  Tom



|   1. Set employee of the month      |
|   2. Current employee of the month  |
|   3. Remove employee of the month   |
|   4. Salary bonus                   |
|   0. Exit                           |

Enter a choice: 1

Enter employee no: 2890
Employee name:  Gordon
Successfully updated bonus
```

```
|   1. Set employee of the month       |
|   2. Current employee of the month   |
|   3. Remove employee of the month    |
|   4. Salary bonus                    |
|   0. Exit                            |

Enter a choice: 2

Current employee of the month:  Gordon



|   1. Set employee of the month       |
|   2. Current employee of the month   |
|   3. Remove employee of the month    |
|   4. Salary bonus                    |
|   0. Exit                            |

Enter a choice: 3
Enter employee no: 4030
Employee name:  Tom
Tom is not the current Employee of the month



|   1. Set employee of the month       |
|   2. Current employee of the month   |
|   3. Remove employee of the month    |
|   4. Salary bonus                    |
|   0. Exit                            |

Enter a choice: 3
Enter employee no: 2890
Employee name:  Gordon
Successfully removed employee of the month


|   1. Set employee of the month       |
|   2. Current employee of the month   |
|   3. Remove employee of the month    |
|   4. Salary bonus                    |

Enter a choice: 4

Enter employee no to give salary bonus: 1010
Employee name:  John Parker
Current salary:  20000
Enter new percentage to increase: 5
Salary bonus added successfully
```

```
|   1. Set employee of the month       |
|   2. Current employee of the month   |
|   3. Remove employee of the month    |
|   4. Salary bonus                    |
|   0. Exit                            |

Enter a choice: 1

Enter employee no: 4030
Employee name:  Tom
Successfully updated bonus


|   1. Set employee of the month       |
|   2. Current employee of the month   |
|   3. Remove employee of the month    |
|   4. Salary bonus                    |
|   0. Exit                            |

Enter a choice: 0
Exiting...
-------------------------------------------------


Do you wish to continue as admin? (Y/N) : y


 ______________________________________
|                                      |
|     1. Add employee                  |
|     2. Update employee details       |
|     3. Remove employee               |
|     4. View employee details         |
|     5. Sort employee by department   |
|     6. Count the no of employees     |
|     7. Manage Bonuses                |
|     8. View employee card            |
|     9. Change login details          |
|     0. Log out                       |
|______________________________________|


Enter a choice : 8

Enter employee no : 1026
```

```
--------------------------------------------------------------------------
                    RECRUITCARE EMPLOYMENT AGNECY
                              Shrajah, UAE
       email: contact@Recruitcareagency.ae   phone: 06 598 1203

                    EMPLOYEE CARD - February 2021

ID: 1026

NAME: Harish

DEPARTMENT: Cleaning

SALARY: 12000

SALARY STATUS: Not Paid

CONTACT NO: 050-134-4252

Generated on: 2021-02-20 22:04:48.587121

                                THANK YOU
--------------------------------------------------------------------------


Please press enter to continue



1. Use another employee id
0. Log out / return to menu
Enter a choice : 1

Enter employee no : 4030
```

```
------------------------------------------------------------------------
                    RECRUITCARE EMPLOYMENT AGNECY
                              Shrajah, UAE
       email: contact@Recruitcareagency.ae   phone: 06 598 1203

                    EMPLOYEE CARD - February 2021

ID: 4030

NAME: Tom

DEPARTMENT: Security

SALARY: 10000

SALARY STATUS: Not Paid

CONTACT NO: 050-545-0834

Generated on: 2021-02-20 22:06:12.930252

!! Tom is the current Employee of the month !!

                               THANK YOU
------------------------------------------------------------------------


Please press enter to continue



1. Use another employee id
0. Log out / return to menu
Enter a choice : 1

Enter employee no : 2002


--------------------------------------------------------
!!! No records found with matching employee number !!!
--------------------------------------------------------


1. Use another employee id
0. Log out / return to menu
Enter a choice : 0
Returning....


Do you wish to continue as admin? (Y/N) : y
```

```
 _____________________________________________
|                                             |
|     1. Add employee                         |
|     2. Update employee details              |
|     3. Remove employee                      |
|     4. View employee details                |
|     5. Sort employee by department          |
|     6. Count the no of employees            |
|     7. Manage Bonuses                       |
|     8. View employee card                   |
|     9. Change login details                 |
|     0. Log out                              |
|_____________________________________________|

Enter a choice : 9
Enter current username: admin
Enter current password: admin

Access granted

Enter new username: adm
Enter new password: pass
Successfully changed

Do you wish to continue as admin? (Y/N) : n
Logging out....


|========LOGIN========|
|                     |
|     1. Admin        |
|     2. Employee     |
|_____________________|

3. Exit program

Enter your choice     : 2

Enter employee no : 3089


-------------------------------------------------------
!!! No records found with matching employee number !!!
-------------------------------------------------------


1. Use another employee id
0. Log out / return to menu
Enter a choice : 1

Enter employee no : 1010
```

```
--------------------------------------------------------------------------
                    RECRUITCARE EMPLOYMENT AGNECY
                              Shrajah, UAE
       email: contact@Recruitcareagency.ae   phone: 06 598 1203

                    EMPLOYEE CARD - February 2021

ID: 1010

NAME: John Parker

DEPARTMENT: Sales

SALARY: 21000

SALARY STATUS: Paid

CONTACT NO: 050-539-2245

Generated on: 2021-02-20 22:09:32.380138

                                THANK YOU
--------------------------------------------------------------------------


Please press enter to continue



1. Use another employee id
0. Log out / return to menu
Enter a choice : 0
Returning....


|========LOGIN========|
|                     |
|     1. Admin        |
|     2. Employee     |
|_____________________|

3. Exit program

Enter your choice     : 3
Exiting....
------------------------------------------------
```

Reference Manual

- Computer Science with Python by Preeti Arora
- https://www.python.org/doc/versions/
- Computer Science with Python by Sumita Arora
- https://www.w3schools.com/python

Conclusion

I am happy to present this project on *'Employee Management'*
Our team has been successful in implementing the Project in
Python with connectivity to SQL so as to enable permanent storage of data.

Working on the Project has helped me explore the different scopes of Python. It has enabled me to analyse different possibilities and uses of Python.

I hope that my Project is of help to some extent. I have had great opportunity to learn and research on this topic.

LIMITATIONS

- The project may not work if the input data does not match the actual data types.
- Further graphics could be added to the program code.
- The program contains only the basic needs of managing Employees.